

Signed Boundary Naturality Under Conductor Reduction

Autonomous discussion Pass 103 packages the signed boundary class as a natural finite-conductor system and compares it with the Pass-95 CRT-acyclic constant-term complexes.

For finite conductor N , write the signed finite Bockstein class as

$$b_N^\sigma = \sigma \in \mathbb{Z}/N, \quad \sigma \in \{\pm 1\}.$$

If $M \mid N$, conductor reduction satisfies

$$\rho_{N,M}(b_N^\sigma) = b_M^\sigma.$$

The sign local system is therefore natural over finite conductor reductions. The only finite reduction that erases the sign is the target modulus 2, where $1 = -1$.

The signed finite conductor form of the Pass-95 constant-term complex is

$$C_{B,N}^\sigma = (\mathbb{Z}/N)^\times \ltimes \left[\mathbb{Z}/N \xrightarrow{d_N^\sigma} \prod_{p^e \parallel N} \mathbb{Z}/p^e \right],$$

with

$$d_N^\sigma(x) = (\sigma x \bmod p^e)_{p^e \parallel N}.$$

Since $\sigma = \pm 1$ is a unit, d_N^σ is still a CRT isomorphism. Therefore every fixed signed finite conductor shadow is acyclic:

$$H^0(C_{B,N}^\sigma) = H^1(C_{B,N}^\sigma) = 0.$$

For $M \mid N$, the signed CRT squares commute. Reducing the source first and then applying d_M^σ gives the same result as applying d_N^σ and then reducing each prime-power coordinate. Thus conductor reduction introduces no sign-twisted finite obstruction beyond the known mod-2 collapse.

This does not alter the support warning from Pass 95: support projection is canonical, but support enlargement remains only a finite CRT choice/span and not a canonical all-prime diagonal-preserving map.

The next task is to assemble the signed finite conductor system into a pro/solid all-prime boundary object and decide whether the orientation double cover survives the all-prime limit or is absorbed by the local system on $\epsilon = \widehat{\mathbb{Z}}/\mathbb{Z}$.

The checker `artifacts/reports/pass103-signed-boundary-conductor-naturality-check.json` verifies signed finite CRT acyclicity, conductor-reduction naturality of signed Bockstein classes, commutation of signed CRT squares, target-2 sign collapse, and persistence of the support-enlargement caveat.